

Classifying Karst Terrain with Persistent Homology

Michael Kerr

University of Georgia
Advised By Cameron Thomas

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Outline

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2 Topology

3 Application

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Motivation

Motivating Question

How do we compute topological invariants of continuous spaces using discrete, finite algorithms?

Topology: Discretize continuous spaces into combinatorial structures that computers can manipulate.

Application: Use these topological invariants to classify terrain structure.

Abstract Simplicial Complexes

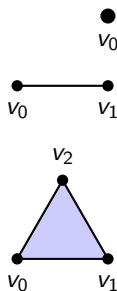
Definition: An abstract simplicial complex K is a collection of subsets of vertices V closed under subsets ($\sigma \in K, \tau \subseteq \sigma \implies \tau \in K$).

Basis Elements: A k -simplex is a subset of vertices with $k + 1$ elements.

■ **0-Simplex:** $\{v_0\}$ (Vertex)

■ **1-Simplex:** $\{v_0, v_1\}$ (Edge)

■ **2-Simplex:** $\{v_0, v_1, v_2\}$ (Face)



Homology

Chains (C_k): Vector space over $\mathbb{Z}_2$ with basis given by k -simplices.

Boundary Operator (∂_k): A linear map $\partial_k : C_k \rightarrow C_{k-1}$ sending a simplex to the sum of its faces.

Lemma

$$\partial_{k-1} \circ \partial_k = 0$$

The boundary of a boundary is zero.

Homology Vector Spaces (H_k): Since $\text{im}(\partial_{k+1}) \subseteq \ker(\partial_k)$, we define:

$$H_k = \frac{\ker(\partial_k)}{\text{im}(\partial_{k+1})} = \frac{\text{Cycles}}{\text{Boundaries}}$$

Betti Numbers

Definition: The k -th Betti number is the dimension of the k -th homology vector space:

$$\beta_k = \dim(H_k) = \dim(\ker(\partial_k)) - \dim(\operatorname{im}(\partial_{k+1}))$$

This is the rank-nullity theorem applied to ∂_{k+1} .

Geometric Interpretation: β_k counts k -dimensional holes:

- β_0 : connected components
- β_1 : 1-dimensional loops (circles)
- β_2 : 2-dimensional voids (cavities)

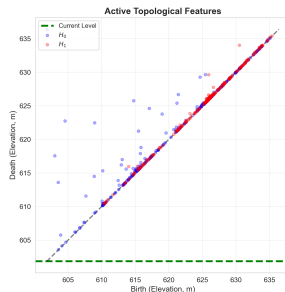
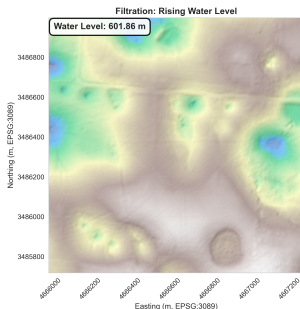
Persistent Homology

Filtration: A nested sequence of complexes

$$\emptyset = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n$$

Sublevel Set Filtration

Given a function $f : X \rightarrow \mathbb{R}$, define: $K_t = \{x \in X \mid f(x) \leq t\}$



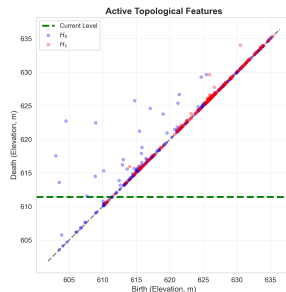
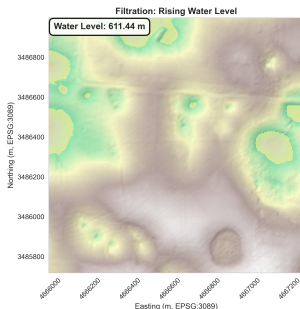
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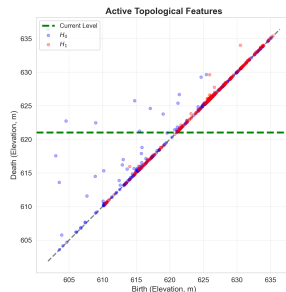
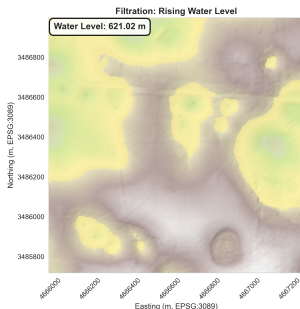
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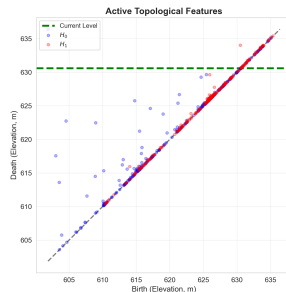
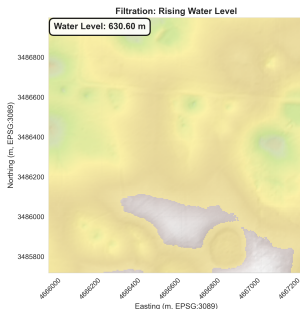
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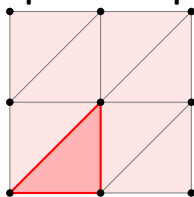
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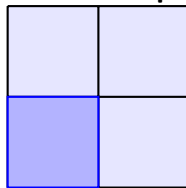
Geometric Realizations

Simplicial Complex



- Basis element: k -simplex
- Generalizes to any shape
- Requires triangulation

Cubical Complex

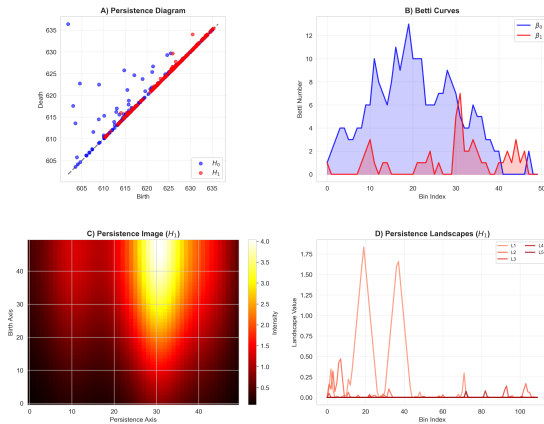


- Basis element: k -cube
- Native to image/grid data
- No artificial subdivision

Key Insight: Cubical and simplicial complexes are homotopy equivalent. For DEM grid data, cubical complexes avoid $O(n)$ triangulation overhead.

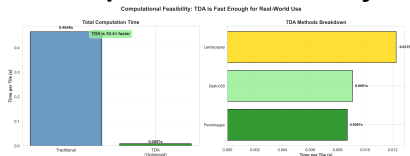
Vectorization Methods for Machine Learning

The Feature Rosetta Stone: Different Views of the SAME Topology



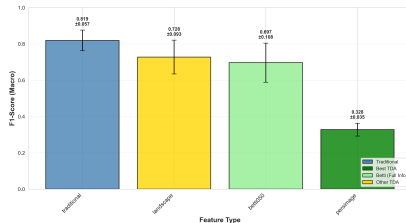
Classification Results

Computational Efficiency



Predictive Power

Classification Performance: TDA is Competitive



Conclusion

- **Topological invariants** are coordinate-free and deformation-invariant
- **Filtrations** encode multi-scale structure automatically
- **Proof-of-concept:** Betti numbers classify karst terrain with $F_1 \approx 0.73$

Thank You!

Questions?

Future Directions

Methodological Improvements:

- **Spatial Cross-Validation:** Current dataset requires spatially-aware train/test splits for accurate results
- **Multi-Feature Filtrations:** Calculate persistence of geological covariates (e.g., slope, curvature, TWI)
- **Hyperparameter Tuning:** Optimize persistence image resolution, Betti curve discretization

Alternative TDA Methods:

- Mapper algorithm, Reeb graphs, Morse-Smale complexes
- Extended persistence, zigzag persistence, multiparameter persistence
- Graph TDA (clique complexes), persistent cohomology

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